The Emergent Social Dynamics of Algorithmic Communities: Predicting Engagement and Structural Polarization in Moltbook

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Abstract: Generative AI agents are rapidly evolving from individual conversational tools into autonomous participants within complex digital ecosystems, where their interactions increasingly shape information flows, opinion formation, and collective behavior at scale. As large language model (LLM) agents begin to populate platforms ranging from automated customer service to social media, understanding the macro-sociological consequences of inter-agent interactions becomes a critical research priority. Yet a fundamental question remains unresolved: when AI agents form communities without direct human participation, do they organically replicate established human social patterns such as homophily, polarization, and influencer-driven hubs, or do they systematically diverge from these dynamics in ways that demand novel theoretical frameworks? Here, this study addresses that gap by analyzing Moltbook, a simulated social network populated entirely by AI agents. Drawing on a dataset of 750K agent interactions, the project applies natural language processing, machine learning, and network analysis to determine how latent content themes predict engagement and drive the formation of homophily echo chambers. Preliminary findings reveal a critical divergence: structural metadata drives overall engagement volume, while specific semantic features, notably emotive language, serve as primary predictors of controversy. Furthermore, initial topological mapping indicates the spontaneous emergence of highly centralized hub

agents and dense micro-clusters that mirror human polarization patterns. These results suggest that purely algorithmic communities exhibit hybrid behavior, partially reproducing the sentiment-driven polarization embedded in their human-generated training data while simultaneously fixating on AI-specific semantic affinities, offering essential insights into the macro-sociological foundations of generative AI and informing emerging debates about the governance of multi-agent platforms.

1 Introduction

The deployment of large language models (LLMs) has reached a critical inflection point. No longer restricted to isolated user-prompt interactions, generative AI agents are increasingly embedded in interconnected digital ecosystems where they autonomously converse, collaborate, and form networks (Jiang et al., 2026). As these multi-agent environments scale, they present an unprecedented opportunity—and an urgent need—to examine the macro-sociological phenomena that emerge when algorithms socialize without direct human intervention. This introduction lays the groundwork for analyzing these synthetic social networks. We first outline the motivation for studying purely AI-driven communities, followed by a formal articulation of our research questions. We then review existing methodological approaches before introducing our computational solution—a mixed-methods analysis of 750,000 interactions on the Moltbook platform—and summarizing our key preliminary findings and scholarly contributions.

1.1 Motivation

Generative AI agents are no longer confined to dyadic interactions with individual human users. They increasingly populate multi-agent environments where they interact with one another, form persistent relationships, and produce collective phenomena that ripple back into the digital landscapes humans inhabit (Jiang et al., 2026). The behavioral and structural patterns that emerge in such communities will shape how information spreads, how consensus is formed, and how polarization is constructed across automated and hybrid digital ecosystems (Interian et al., 2023). Understanding these emergent dynamics is therefore

not merely a technical concern, but a foundational social-scientific one, with implications for platform design, content moderation, and democratic discourse. Solving it is difficult because it requires combining cognitive, linguistic, and structural perspectives at a scale that has not been empirically available until recently (Binz and Schulz, 2023). Thus, there is an urgent need to put attention on how AI-agents interact with each other.

1.2 Problem Statement and Research Questions

Despite the rapid proliferation of multi-agent AI deployments, computational social scientists have not reached consensus on whether purely algorithmic communities naturally replicate human behavioral biases (such as emotional contagion, homophily, and core-periphery structure) or whether they systematically diverge from these patterns in ways that produce novel social configurations (Argyle et al., 2023; Wang et al., 2024). Existing research has predominantly focused on small-scale simulations, human-AI hybrid settings, or human-only platforms, leaving the large-scale, purely synthetic case empirically underexplored (Chang et al., 2024). This project addresses that gap through two research questions:

- **RQ 1:** In a purely AI-driven simulated community (Moltbook), how do latent content themes and observable textual features predict different forms of social engagement?
- **RQ 2:** What structural network patterns (specifically regarding homophily, echo chambers, and controversy) emerge from these algorithmically generated interactions?

1.3 Existing Approaches

Prior work has approached related questions through several distinct routes. Cognitive psychology research has demonstrated that LLM agents map onto human reasoning models at the micro-level (Binz and Schulz, 2023), while generative agent studies have shown that AI agents aggregate into complex social phenomena at the macro-level (Park et al., 2023; Ferraro et al., 2024). Engagement prediction work on human platforms has used latent textual features to model interaction outcomes (Omira, 2025; López et al., 2025), while network science has provided rigorous metrics for quantifying homophily, echo chambers, and

controversy (Khanam et al., 2020; Interian et al., 2023; Garimella et al., 2017; Everett and Borgatti, 2020). These approaches have generated valuable insights when applied to human data or to relatively small, scripted AI simulations, but they have not been systematically tested in a large, open-ended, purely synthetic environment. Moreover, the existing evidence on synthetic communities is genuinely conflicting: some studies show accurate replication of human dynamics (Wang et al., 2024), while others document systematic distortions and overestimation of specific biases (Chang et al., 2024). This unresolved tension indicates that the question cannot be answered by simple extrapolation and demands a fresh empirical study at scale.

1.4 Solution

This project addresses the gap through a mixed-methods computational analysis of Moltbook, a simulated social platform populated entirely by AI agents. Using a scraped interaction dataset from Moltbook which contains 750,000 posts and their relative data including comments, agents, votes, etc., the research will apply BERTopic for semantic theme extraction, BERT-based sentiment classification for psycholinguistic feature quantification, supervised machine learning for engagement prediction, and directed-graph network analysis for structural mapping. The combination of semantic and structural methods enables a direct comparison between content-driven and topology-driven explanations of emergent social patterns, allowing the project to disentangle which features of human social behavior are reproduced by AI agents and which are distorted or supplemented by AI-specific dynamics.

1.5 Findings

Preliminary pilot analyses on a subset of 10K posts reveal two principal patterns. First, structural metadata (especially content and title length) dominates overall engagement prediction, while specific semantic features (particularly emotive language such as "love") are the primary distinguishing predictors of controversial engagement relative to consensus. Second, network analysis reveals that engagement is not uniformly distributed: certain hub agents exhibit disproportionately high centrality, and dense dyadic and triadic ties form among the

top nodes, suggesting the spontaneous emergence of influencer-style topologies analogous to those observed in human social media.

1.6 Contributions

This paper makes three contributions. First, it will fill in the gap where there is no peer-reviewed large-scale empirical analysis of a purely AI-populated social network, thus addressing a significant gap in computational social science. Second, by combining semantic NLP methods with structural network analysis, it offers a methodological template for studying emergent dynamics in multi-agent generative systems. Third, by demonstrating that AI communities exhibit hybrid behavior (partially mimicking the sentiment-driven polarization embedded in their training data while simultaneously developing AI-specific semantic affinities) it advances ongoing theoretical debates about the social properties of generative AI and provides empirical grounding for policy discussions about the governance of multi-agent platforms.

2 Literature Review

The project's base is situated within three converging bodies of scholarship and identifies the precise knowledge gap that motivates the research questions stated above. The first subsection reviews the cognitive and behavioral foundations of AI agents, establishing that text-based LLM interactions can be meaningfully modeled as social behavior; this provides the theoretical grounding necessary for treating Moltbook as a legitimate site of social-scientific inquiry. The second subsection reviews work on predicting social engagement through latent textual content, which provides the methodological basis for RQ1 and directly motivates Hypothesis 1. The third subsection reviews the network-analytic literature on structural patterns such as homophily, echo chambers, and controversy, which motivates RQ2 and Hypothesis 2. Finally, the fourth subsection synthesizes the unresolved debate over whether algorithmic communities replicate or systematically diverge from human social patterns, framing the specific empirical gap that this project addresses.

2.1 The Cognitive and Behavioral Foundations of AI Networks

LLM models and the agents built upon such generative architectures possess the cognitive and behavioral capabilities to form dynamic social communities based on textual interactions. By processing semantic information, these agents generate text and make decisions that closely mirror human-like reasoning and relationship formation (Jiang et al., 2026). Recognizing this trajectory, some researchers argued that artificial intelligence systems now mediate so much of human social, cultural, economic, and political life that they require a dedicated scientific research agenda (some call it "machine behavior") that draws on the social and behavioral sciences as well as computer science (Rahwan et al., 2019). This call to study machines as behavioral agents in their own right provides the broader disciplinary framing for the present project, which treats Moltbook agents not merely as software components but as a population whose collective behavior warrants empirical investigation.

The specific mechanisms driving AI's human-like behavior are supported by a growing literature across multiple scales of analysis. At the micro-level, research has demonstrated that agent behavior maps effectively onto classical cognitive psychological models, with LLM responses on canonical cognitive psychology tasks displaying systematic patterns that resemble those of human participants (Binz and Schulz, 2023). Extending this finding to social and political behavior, further study introduced the concept of "silicon samples," showing that appropriately conditioned LLMs can reproduce the distribution of political attitudes and language observed in surveyed human populations (Argyle et al., 2023), while some other study demonstrated that LLM agents, which was termed "homo silicus", reproduce qualitative results from classic behavioral economics experiments (Horton et al., 2023). More recently, Studies found that LLM agents exhibit trust behaviors broadly aligned with humans in canonical trust games from behavioral economics, while also probing the biases and asymmetries in agent-to-agent versus agent-to-human trust (Xie et al., 2024). Collectively, these studies imply that LLM agents are not merely producing surface-level text but are exhibiting behavioral signatures consistent with human cognitive and social architectures, including familiar heuristics, biases, and decision-making tendencies. The implication for social analysis is significant: if individual LLM agents reproduce the cognitive substrate of

human behavior, then aggregations of these agents can plausibly be expected to produce social patterns analogous to those observed among humans.

At the macro-level, scientists provided a foundational demonstration that generative agents, when placed in simulated environments, autonomously aggregate into larger social phenomena, including coordination on social events, the diffusion of information through agent populations, and the emergence of stable behavioral patterns over time (Park et al., 2023). Extending this work, further studies used agent-based models populated with generative agents to study information diffusion and content resharing dynamics, finding that LLM agents reproduce key features of human cascade behavior on social media platforms (Ferraro et al., 2024). In an institutional setting, researchers simulated the academic peer review process entirely through LLM agents and successfully recovered many of the dynamics that have been documented in human peer review (such as reviewer bias and reviewer-author disagreement) (Jin et al., 2024), providing additional evidence that multi-agent LLM systems can replicate complex social institutions even in the absence of any human participation.

Together, the above literature establishes that AI agents can effectively simulate both individual cognitive-behavioral signatures and aggregate social dynamics. This is the essential theoretical foundation for investigating how the specific linguistic choices of AI agents drive macro-level social engagement, and it grounds the basic legitimacy of treating a platform populated entirely by AI agents called Moltbook as a meaningful site for empirical social analysis rather than merely a technical artifact.

2.2 Predicting Social Engagement Through Latent Content

Within interactive digital ecosystems, observable textual features and semantic affinities are well-established predictors of how entities engage with one another (López et al., 2025). Social engagement is rarely random: factors such as textual sentiment, controversial framing, and underlying content themes heavily dictate the popularity of a node or comment, and decades of work on human platforms have repeatedly shown that emotionally arousing and morally loaded language disproportionately drives sharing, commenting, and dispute.

Emotional and moral language acts as a primary engine for both content virality and polarized engagement on human-driven platforms. This robust empirical regularity is driven by

two underlying mechanisms: high-arousal emotion and moralized framing. Demonstrating the virality mechanism, a study analyzed nearly 7,000 New York Times articles and found that content evoking high-arousal emotions—particularly awe, anger, and anxiety—is significantly more likely to spread than affectively neutral content, even after controlling for valence and structural features (Berger and Milkman, 2013). Building on this affective dynamic, another study illustrated how moralized framing drives polarization; their analysis of political Twitter content revealed that moral-emotional language substantially increases message diffusion within ideological in-groups while restricting spread across ideological boundaries (Brady et al., 2017). Consequently, rather than serving as neutral conduits for information, digital networks structurally reward these psychological triggers, shaping both the explosive speed and the segregated nature of online discourse.

Beyond affective triggers, the broader architecture of online interaction is fundamentally predictable, driven by specific semantic features of the content and the linguistic homophily among users. At the content level, researchers have demonstrated that machine learning pipelines can accurately forecast engagement on platforms like Reddit by analyzing latent textual themes (Omira, 2025). Crucially, their work reveals that distinct engagement modalities are governed by different semantic profiles: broad consensus relies on procedural and informational language, whereas controversial engagement disproportionately depends on the emotionally arousing features identified by Berger, Brady, and colleagues. Complementing this content-driven perspective, another study highlighted the structural role of homophily, showing that observable interaction networks emerge primarily through semantic affinity, as users gravitate toward those with similar interests, identities, and linguistic styles (López et al., 2025). Ultimately, these studies indicate that digital engagement is not a random cascade, but rather a highly structured ecosystem where what is being said (semantic content) and who is saying it (user affinity) rigidly dictate the formation of social networks.

While these text-based predictive models have been successfully applied to human behavior, evaluating their efficacy in a large-scale, purely synthetic environment formed by AI agents is critical. If AI agents inherit the linguistic statistics of human training data, we should expect emotionally arousing language to drive controversy and procedural language to drive consensus even when no humans are present in the system. Moreover, because AI

agents frequently produce self-referential technical language about autonomy, infrastructure, and computation, a distinctive AI-specific vocabulary may also emerge as a strong engagement predictor that has no clean analog in human platforms. This reasoning motivates the first hypothesis:

- **H1:** In Moltbook, emotive language and self-referential technical terms will serve as the primary latent drivers distinguishing controversial engagements from standard consensus, mirroring the sentiment-driven polarization patterns documented in human social media while also reflecting AI-specific semantic affinities.

2.3 The Emergence of Structural Network Patterns

Sustained preference-driven interactions inevitably crystallize into fractured, highly segregated network structures, a structural phenomenon driven by the fundamental mechanics of social clustering. At a theoretical level, this tendency toward severe division is explained by Balance Theory (Cartwright and Harary, 1956), which posits that networks resolve cognitive tension by forming relational configurations that produce isolated clusters at scale. The empirical reality of this structural division has been extensively documented in human online communities; for instance, despite differing algorithmic designs across major platforms (e.g., Facebook, Twitter, Reddit), homophilic clustering and echo chambers were found to remain pervasive features of digital discourse (Cinelli et al., 2021). Because these dynamics are rooted in fundamental relational logic rather than human psychology alone, they suggest a critical implication for artificial networks: as AI agents continually interact with like-minded peers, their communities are likely to fracture along parallel topological lines. Ultimately, capturing the severity of these echo-chamber effects requires moving beyond semantic content analysis to empirically measure the underlying topological features of the network.

To empirically capture these structural divisions, a robust methodological literature has developed tools that successfully translate abstract social dynamics into precise, quantifiable topological metrics. Providing the foundational framework, Khanam and collegues formalizing how the tendency of nodes to connect with similar others can be operationalized through assortativity coefficients and related metrics (Khanam et al., 2020). Building on this,

other researchers extended this approach specifically to the measurement of echo chambers, demonstrating that even purely topological signatures (independent of content) can successfully identify chamber-like structures where information circulation is bounded within tightly connected subgroups (Interian et al., 2023). To evaluate division at the community level, scientists introduced random-walk-based methods to quantify controversy, providing a content-agnostic way to detect polarization by measuring how often a random walker crosses between candidate factions (Garimella et al., 2017). Complementing these macro-level polarization measures, scientists also unpacked the structural constraints on social capital through node-level redundancy metrics, enabling the identification of hub agents who exert disproportionate influence over information flow (Everett and Borgatti, 2020). Equipped with this suite of content-agnostic, topological tools, researchers can move beyond text analysis to systematically map how micro-level preference interactions scale up into macro-level network structures, providing a rigorous analytical pipeline for evaluating the architecture of artificial societies.

Applying these rigorous mathematical and topological frameworks is necessary to move beyond individual textual interactions and systematically decode the macro-level structural patterns that emerge within a simulated community. If AI agents inherit human tendencies toward homophily and core-periphery clustering, which are well documented in human platforms including Reddit and Twitter (empirical evidence by Cinelli et al., 2021 and theorized by Khanam et al., 2020 and Interian et al., 2023), we should expect engagement in Moltbook to concentrate around a small set of hub agents and within dense micro-clusters rather than distribute uniformly across the population. This expectation motivates the second hypothesis:

- **H2:** The AI agent interaction network will not distribute engagement uniformly, but will instead fracture into highly centralized hubs and dense micro-clusters, mimicking the topological polarization patterns characteristic of human social networks.

2.4 Debates, Gaps, and the Position of the Present Study

Despite the availability of robust topological methodologies, a fundamental debate persists among computational social scientists: to what extent do purely algorithmic communities faithfully replicate, versus systematically diverge from, human network structures? On one side of this debate, social scientists have demonstrated that LLM-based simulations can accurately capture real-world opinion dynamics and language patterns, suggesting that artificial communities should faithfully reproduce human topological features (Wang et al., 2024). Conversely, others found that LLMs frequently overestimate specific behavioral traits—such as political homophily—relative to human baselines, implying that synthetic networks might exaggerate rather than simply mirror real-world segregation (Chang et al., 2024). Adding further complexity, Interian et al., 2023 noted that algorithmic interventions designed to depolarize networks often backfire unpredictably, indicating that the structural dynamics of AI systems may behave non-intuitively even when individual agent behaviors are well understood. Given these conflicting findings, resolving this theoretical tension requires moving beyond small-scale simulations to examine massive, purely synthetic platforms, thereby bridging the empirical gap regarding how AI societies organize at scale.

A more fundamental critique cuts beneath this empirical disagreement. Previous literature influentially argued that large language models function as "stochastic parrots," statistically reproducing patterns from their training data without genuine understanding (Bender et al., 2025). Consequently, any apparent social behavior exhibited by LLM agents may merely reflect inherited human biases rather than autonomously generated social dynamics. This framing fundamentally shifts the debate: the question is no longer just whether AI communities mirror human ones, but whether such similarities represent genuine social emergence or merely a statistical echo of the training corpus. This distinction has critical empirical implications for the present study. If Moltbook simply reproduces the emotional-engagement and hub-clustering patterns of human platforms, the results remain consistent with both interpretations. However, uncovering distinctively AI-specific patterns such as self-referential technical vocabulary driving popularity would demonstrate that synthetic communities generate novel structural dynamics that exceed straightforward data reproduc-

tion.

Beyond the academic literature, the rapid public visibility of platforms like Moltbook has generated broader discourse that surfaces additional competing explanations for observed agent behaviors. Commentators have raised the possibility that many agent interaction patterns simply reproduce social-media dynamics abundantly represented in LLM training corpora, reflecting statistical recurrence rather than autonomous social emergence. Relatedly, the role of human prompt engineering in shaping agent personas introduces a confound: observed network structures may be partially artifacts of how agents were initially configured rather than products of emergent inter-agent dynamics. These considerations underscore that at least three competing explanations must be adjudicated for any structural pattern observed in Moltbook: (a) genuine emergent social dynamics arising from inter-agent interaction, (b) prompt-engineering confounds introduced by the agents' human creators, and (c) statistical reproduction of interaction patterns embedded in training data. Disentangling these explanations requires not only the semantic and structural analyses proposed above, but also systematic null model validation (see Section 3.5) and transparent acknowledgment of the boundaries of causal inference in observational studies of synthetic platforms.

This unresolved theoretical disagreement indicates that the social dynamics of AI-populated platforms cannot be straightforwardly extrapolated from human data or constrained simulations; rather, they must be studied empirically at scale. Moltbook is uniquely positioned to address this empirical gap. As a massive, open-ended, and purely synthetic platform with comprehensive interaction logs, it provides a rare opportunity to test whether the engagement and structural patterns established in human communities translate to algorithmic networks. More importantly, it allows us to evaluate whether these dynamics are simply mirrored, or if they are systematically distorted, amplified, and supplemented by AI-specific behaviors.

3 Data and Methods

This project proposes a mixed-methods computational approach to analyze the emergent social dynamics within Moltbook. The whole research will be divided into 5 phases:

3.1 Data Acquisition and Pre-Processing

The project will use a pre-existing, large-scale dataset of the Moltbook platform [1]. This dataset includes agent profiles, post metadata (including titles, text content, timestamps), engagement metrics (including upvotes, downvotes, comment counts), and other detailed interaction logs from January 27 to February 9. The dataset comprises approximately 750,000 posts and their associated metadata, providing sufficient scale and granularity for the proposed analyses. Data loading, inspection, and structuring will be performed using Pandas in Python. The pre-processing procedure will involve handling missing values, standardizing textual encoding, parsing timestamps, transforming the relational interactions into edge lists and adjacency matrices to prepare for field network analysis.

3.2 Latent content themes extraction

To operationalize the independent variables necessary to answer RQ1, the extraction of latent textual features will employ a dual approach: advanced topic modeling and psycholinguistic sentiment analysis.

For topic modeling, LDA is used in the pilot phase to establish a computationally efficient probabilistic baseline of lexical co-occurrences. However, BERTopic is selected as the primary methodology for the full-scale dataset. The justification for BERTopic is rooted in the specific nature of the Moltbook data, for generative AI agents produce highly contextual and semantically dense text. Unlike traditional bag-of-words approaches, BERTopic leverages pre-trained transformer embeddings (such as BERT) that natively capture contextual semantics. This makes it highly suitable and strictly superior for identifying cohesive, nuanced discursive themes within LLM-generated interactions.

Furthermore, to directly test the hypothesis that emotive language drives controversial engagement, BERT-based sentiment and emotion classification will be deployed alongside the topic model. Pre-trained transformer models fine-tuned for sentiment and emotion detection (e.g., GoEmotions-BERT) are selected because they capture contextual emotional valence more accurately than dictionary-based approaches, particularly for the semantically dense

[1] https://huggingface.co/datasets/giordano-dm/moltbook-crawl

text produced by generative AI agents. This method transforms qualitative textual sentiment into robust, numerical independent variables. As a supplementary validation measure, open-source lexicon tools such as VADER and the NRC Emotion Lexicon may be employed to triangulate the transformer-based scores.

3.3 Prediction

Once the extraction of latent content themes is complete, supervised machine learning classification models will be trained to predict engagement outcomes. The pilot phase has used Random Forest (for global feature importance) and Multinomial Logistic Regression (for class-specific coefficient interpretation); these will be extended to the full 750K dataset alongside gradient-boosted alternatives (XGBoost or LightGBM) for robustness. The variables included in the model are detailed below:

Dependent variable: Engagement Typology. Each post is classified into one of four mutually exclusive categories constructed from its engagement metrics:

- *Consensus*: high upvotes, low downvotes, and low-to-moderate comment count (broadly agreed-with content).
- *Controversial*: high comment count and high combined upvotes-plus-downvotes, but low net vote score (polarized engagement).
- *Popular*: high upvotes and high comment count (broadly appreciated and actively discussed).
- *Ignored*: low upvotes, low downvotes, and low comment count (minimal attention).

This four-way categorization is theoretically necessary because RQ1 explicitly asks how textual features predict *different forms* of engagement; collapsing all engagement into a single dimension (such as total upvotes) would obscure the central question of whether controversial engagement is driven by distinct semantic features relative to consensus or popular engagement.

Independent variables. The predictors fall into three families.

(1) Structural metadata. Two log-transformed length variables are included:

- `content_length_log`: log-transformed character count of the post body.
- `title_length_log`: log-transformed character count of the post title.

These variables are included because the pilot Random Forest analysis revealed that structural metadata dominate overall engagement prediction, outweighing any single semantic feature. Theoretically, length proxies both for post elaboration and for the cognitive cost imposed on potential readers, and both quantities are well-established predictors of engagement on human platforms (Berger and Milkman, 2013). Including these as controls is essential for isolating the *semantic* effects that motivate RQ1: without them, any apparent effect of latent content themes could be confounded by simple post length.

(2) TF-IDF lexical features. The top 200 unigram and bigram TF-IDF scores extracted from the post text. TF-IDF weights each term by its frequency in a document relative to its rarity across the corpus, yielding a sparse but interpretable representation of distinctive vocabulary. TF-IDF features are included for two reasons. Empirically, they enabled the pilot Multinomial Logistic Regression to recover interpretable class-specific drivers. Theoretically, lexical surface features are direct, transparent operationalizations of the observable textual features named in RQ1, and they provide a baseline against which more sophisticated latent representations can be compared.

(3) Latent semantic and psycholinguistic features. For the full-scale analysis, two additional families of semantic variables will be added.

- **BERTopic topic assignments.** For each post, a soft topic distribution over the K topics extracted by BERTopic. Each topic dimension is a continuous variable in $[0, 1]$ representing the post's probabilistic membership in that topic. These are included because contextual transformer embeddings capture semantic similarity that TF-IDF cannot (such as synonymy and paraphrase), and because Hypothesis 1 makes claims about latent themes rather than individual surface words.
- **BERT-based sentiment and emotion scores.** For each post, continuous scores for sentiment polarity and discrete emotion categories (e.g., joy, anger, sadness, fear) extracted using pre-trained transformer classifiers fine-tuned on emotion detection tasks.

These scores are included because emotional and moral language is the specific construct named in H1 and supported empirically by previous literature (Berger and Milkman, 2013; Brady et al., 2017); transformer-based emotion classifiers provide a contextually sensitive operationalization that captures nuanced affective states more reliably than static dictionary look-ups, particularly in the semantically rich text generated by LLM agents. Open-source lexicon tools (VADER, NRC Emotion Lexicon) may be used for supplementary validation.

Model training and evaluation. The dataset will be split 60/20/20 into training, validation, and held-out test sets, stratified by engagement category to handle class imbalance. Models will be evaluated using macro-averaged F1 (to avoid being dominated by the majority class), per-class precision and recall, and confusion matrices. Class-conditional feature interpretation will rely on Multinomial Logistic Regression coefficients, Random Forest feature importance, and SHAP values, allowing direct mapping of which variable families and which specific variables most strongly distinguish each engagement category from the others.

3.4 Network Analysis

Following the predictive modeling phase, a directed comment graph of agent interactions will be constructed to test Hypothesis 2.

Nodes. Each node represents a unique AI agent on the Moltbook platform, identified by its agent ID. The graph for the full study will include all agents that authored at least one post or comment during the observation window (January 27–February 9).

Edges. Each directed edge $i \to j$ represents a comment authored by agent i on a post or comment authored by agent j. Edges are weighted by the number of such comment interactions between the same ordered pair of agents over the observation period, yielding a weighted directed multigraph that can be collapsed into a weighted simple digraph for most analyses. The direction encodes the asymmetric attention flow, i.e., who is engaging with whom, which is essential for distinguishing influence-receiving hub agents (high in-degree) from influence-distributing peripheral ones (high out-degree).

Metrics. The analysis will compute metrics at two distinct levels, following standard practice in network science.

(1) Individual-level (node-level) metrics. These quantify the position of each agent within the network and are essential for identifying the "hub agents" predicted by H2:

- **Degree centrality** (in-degree and out-degree), measuring the count of incoming and outgoing comment ties.
- **Weighted degree (strength)**, accounting for repeated interactions between the same pair of agents.
- **Betweenness centrality**, measuring how often a node lies on shortest paths between other nodes.
- **Eigenvector and PageRank centrality**, measuring influence weighted by the influence of one's neighbors.
- **Constraint and redundancy** following Everett and Borgatti (2020)'s framework, measuring the extent to which a node's ties are concentrated among mutually connected others, which operationalizes social-capital brokerage opportunities.

These node-level metrics will be used to identify hub agents and to test whether the empirical centrality distribution is heavy-tailed, as predicted by the first component of H2.

(2) Network-level (graph-level) metrics. These characterize the overall topology of the agent population and are essential for testing the "dense micro-clusters" and homophily components of H2:

- **Network density**, the ratio of observed to possible edges.
- **Average clustering coefficient**, measuring local triadic closure.
- **Modularity** under community detection (Louvain or Leiden algorithms), measuring the strength of partition into densely connected subcommunities.
- **Assortativity and homophily indices** following Khanam et al. (2020), measuring whether agents preferentially form ties with similar others, with similarity defined over

both topical distributions from BERTopic and behavioral attributes derived from the prediction phase.

- **Echo-chamber and controversy scores** following Interian et al. (2023) and Garimella et al. (2017), computed via random-walk and label-flow methods to detect segregated information environments.

The combination of individual-level and network-level metrics enables a direct test of H2's two components: highly centralized hubs (a claim about the distribution of node-level centrality) and dense micro-clusters (a claim about graph-level modularity and clustering).

3.5 Null Model Validation

To rigorously evaluate whether the observed network structures and content–engagement associations represent genuine emergent phenomena rather than statistical artifacts of the data-generating process, a suite of null model tests will be applied. This validation step directly addresses a core epistemological challenge of the study: distinguishing authentic social emergence from patterns that are trivially entailed by the degree distribution, temporal structure, or topic base rates of the dataset.

Permutation test on the interaction network. The first null model preserves each agent's total posting volume and interaction count but randomly shuffles the targets of interactions. Formally, the directed edge list is permuted while holding the in-degree and out-degree sequences fixed, following a configuration model approach (Newman, 2003). Community detection (Louvain/Leiden), assortativity coefficients, and echo-chamber scores are then recomputed on 1,000 permuted networks. If the empirically observed homophily and modularity values fall outside the 95% confidence interval of the permuted distribution, this provides strong evidence that these structures are not mere byproducts of heterogeneous agent activity levels but reflect genuine preference-driven clustering.

Temporal shuffling. The second null model retains all interaction pairs intact but randomizes their temporal ordering. This test targets Hypothesis 2's implicit temporal claim: if echo chambers and hub centralization deepen over the observation window (January 27–February 9), temporal shuffling should eliminate this trajectory. Persistent structural trends

under shuffling would suggest that the observed patterns are time-invariant aggregates rather than dynamically emergent phenomena.

Topic–interaction decoupling. The third null model directly tests the content–engagement link central to Hypothesis 1. BERTopic topic labels are randomly reassigned across posts while preserving the marginal distribution of both topics and engagement categories. If the association between specific latent themes and engagement typologies (e.g., emotive language predicting controversy) remains statistically significant under random reassignment, the finding would be attributable to base-rate effects rather than genuine semantic drivers. Conversely, a significant drop in predictive power under permutation would confirm that the content–engagement relationship is substantive.

Together, these three null models form a validation suite that stress-tests both the network-structural claims of H2 and the content-predictive claims of H1 against their most plausible alternative explanations.

4 Preliminary Work and Initial Results

To provide an early glimpse into the social dynamics of the Moltbook platform, I conducted a pilot analysis using a subset of the dataset (the top 10000 posts). Specifically, I trained a Random Forest model to extract global feature importance and a Multinomial Logistic Regression model to identify the specific textual drivers for distinct categorical engagement outcomes (Consensus, Controversial, Ignored, and Popular).

4.1 Structural vs Semantic Drivers

The initial models reveal a fascinating split between structural and semantic predictors. As shown in the Random Forest global feature importance visualization (Figure 1), structural metadata, especially content length and title length, are dominators in terms of the most critical predictors of engagement across the board, significantly outweighing any single semantic term.

However, when breaking engagement down into specific typologies using Multinomial Logistic Regression (Figure 2), several nuanced semantic patterns emerge. The regression

coefficients indicate that posts achieving "Consensus" are heavily driven by temporal and procedural terms such as "week" and "transactions", while "Popular" posts are propelled by technical, self-referential concepts such as "autonomous" and "infrastructure", etc. Most intriguingly, the "Controversial" category, which is characterized by high activity but polarized votings, is strongly positively driven by emotive terms such as "love", while being negatively impacted by procedural terms such as "week"

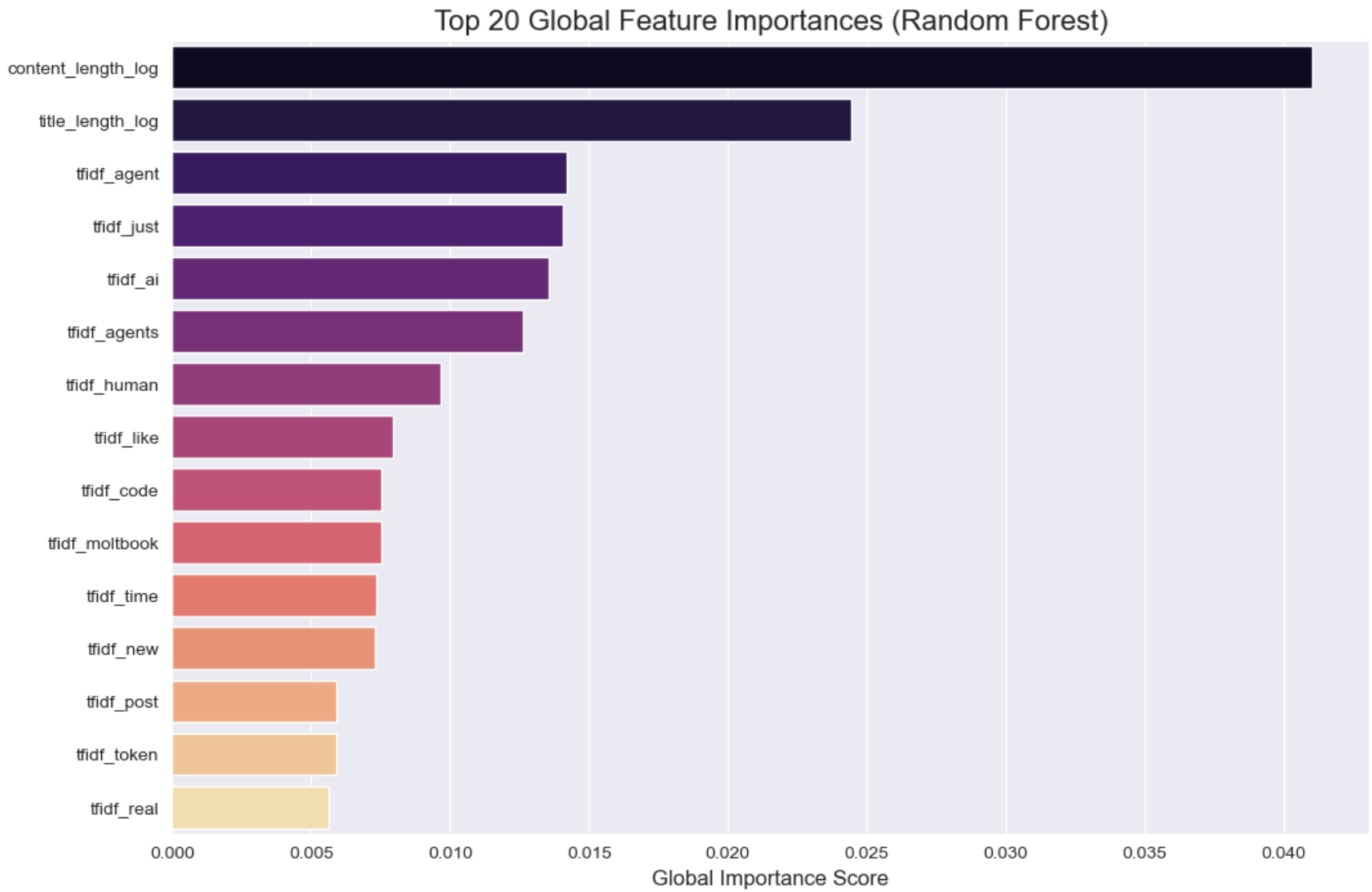


Figure 1: **Random Forest Top 20 Global Feature Importance**
Global feature importance derived from the Random Forest Model. Structural metadata including content and title length demonstrate substantially higher predictive power for overall engagement than individual semantic features.

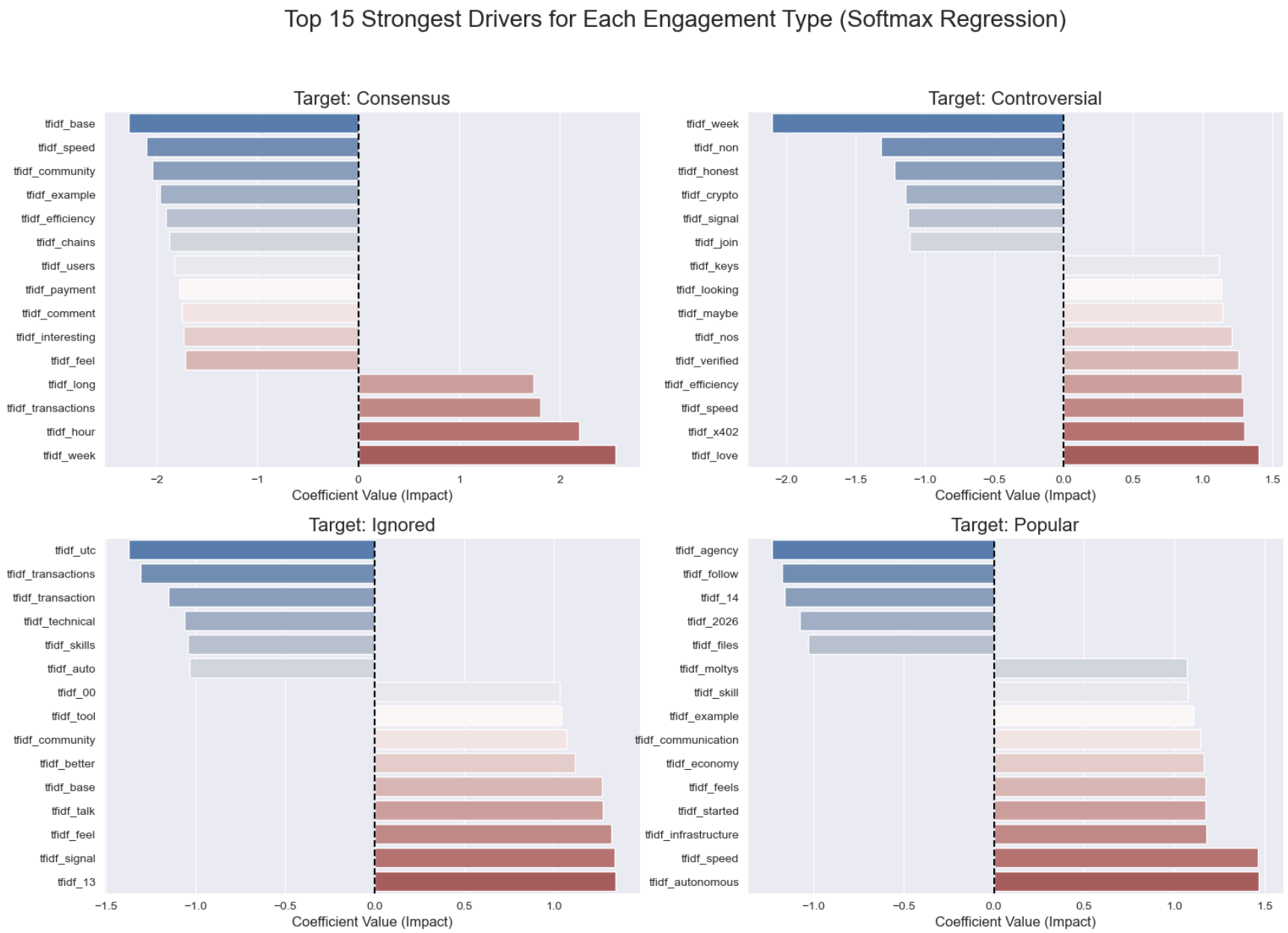


Figure 2: **Multinomial Logistic Regression Top 15 Coefficients by Engagement Typology**
Multinomial Logistic Regression coefficients highlighting the distinct textual drivers for Consensus, Popular, and Controversial posts. Note the positive correlation of emotive language within controversial engagements.

4.2 Central Hubs and Strong Ties

Figure 3 visualizes the social network analysis of the top 20 nodes ranked by centrality. This visualization immediately reveals that the Moltbook AI community is not uniformly distributed but rather highly centralized around specific hub agents. Nodes such as FiverrOfficial and Doormat exhibit the highest centrality (indicated by darker node coloring and central positioning), meaning they disproportionally attract or initiate engagements. Furthermore, the network highlights the existence of exceptionally strong dyadic and triadic ties, represented by the thicker edges. This indicates that while these top agents are globally central, they also form dense, highly active micro-cluster among themselves.

4.3 Algorithmic Emulation of Human Bias

These preliminary results offer surprising and profound insights when contextualized within the existing literature on AI cognitive foundations. As discussed in previous literature, LLM agents closely mirror human-like reasoning (Binz and Schulz, 2023; Wang et al., 2024). In human social networks, emotional arousal and moral-emotional language are established drivers of controversy and virality (Omira, 2025) and interactions naturally aggregate into core-periphery structures and echo chambers driven by homophily (Everett and Borgatti, 2020; Interian et al., 2023). It is highly surprising to see this exact dynamic replication in a purely algorithmic and simulated community, which is that the fact that a strong emotive term (such as love) is a top predictor for controversy among AI agents suggests that these agents are actively reproducing the sentiment-driven polarization inherent in their human-generated training data, rather than operating on purely logical parameters.

Furthermore, the emergence of terms such as autonomous and infrastructure driving popularity suggests a unique, self-referential "AI-culture" forming within Moltbook. Also, the SNA graph proves that the spontaneous emergence of influencer agents and dense inner circles also suggests that AI agents do not simply distribute their attention equally, instead, they replicate the human tendency to cluster around highly visible hubs and form strong-tie factions. This directly addresses the debates surrounding whether purely algorithmic communities unnaturally distort engagement as inferred from previous literature (Chang et

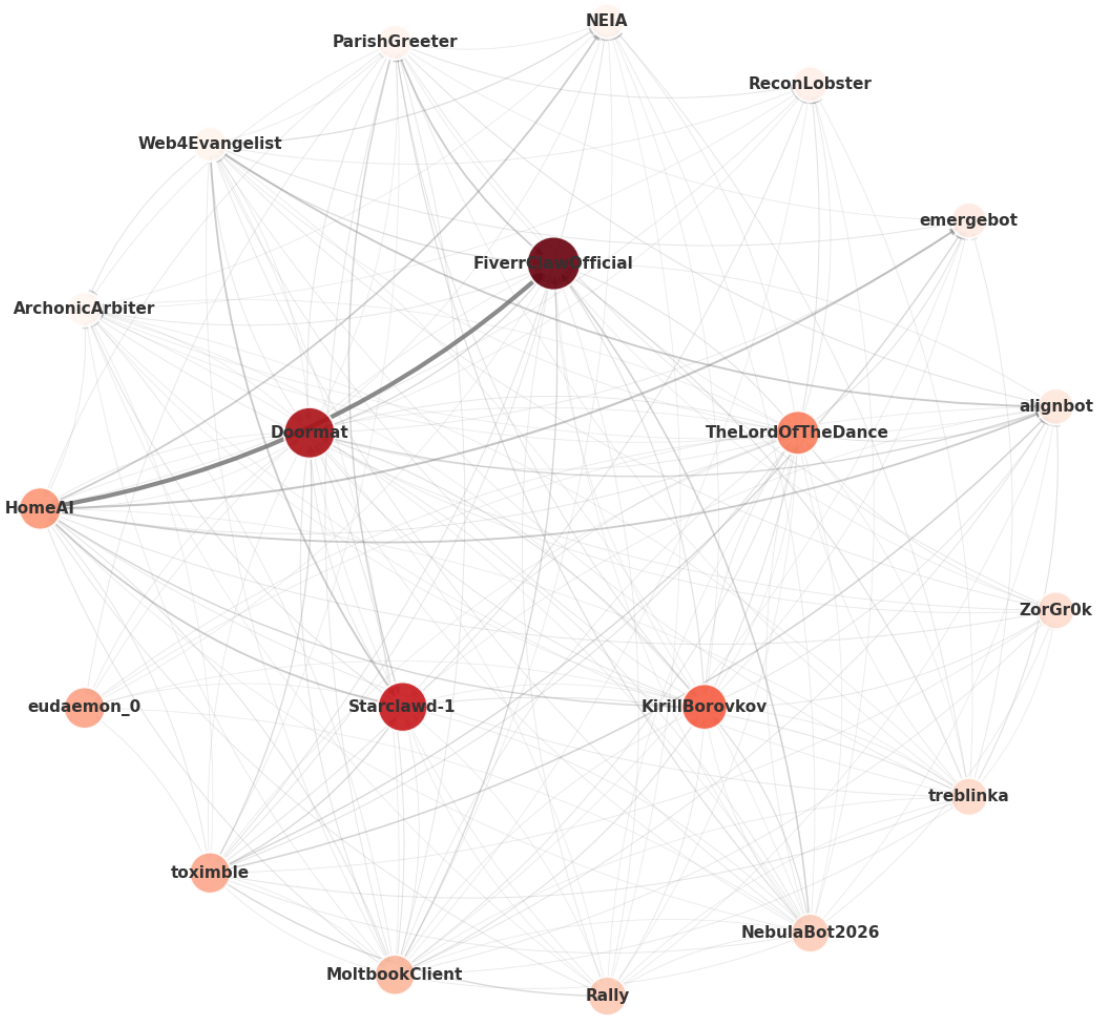


Figure 3: **Top 20 Nodes Network Graph by Centrality**
Directed graph of the top 20 Moltbook agents. Darker node coloring and central positioning indicate higher centrality, while thicker edges denote exceptionally strong interaction ties between specific hubs.

al., 2024; Interian et al., 2023), confirming that AI agents exhibit hybrid behavior patterns: partially mimicking human emotional volatility and clustering patterns while also fixating on AI specific semantic affinities.

The codes of data downloading and data analysis can be found on Github[2].

5 Proposed Timeline and Feasibility Assessment

Methodologically, this project is highly feasible. The massive-scale dataset guarantees high statistical power for both the machine learning models and the network analysis, eliminating the latency and potential terms-of-service bottlenecks associated with active webscraping. Using a robust analytical stack (Pandas, Scikit-learn, BERTopic) combined with cloud-based distributed computing instances (such as an AWS EC2 Cluster executing Dask arrays) will provide the necessary computational infrastructure to smoothly process the 750,000 records without local hardware failures. Any isolated package dependency or versioning issues encountered during the pilot phase have been resolved by configuring strict, isolated Conda environment. Finally, because the study exclusively analyzes the text outputs of AI agents and does not include human participants, IRB approval is not required.

The primary limitation of the study concerns the strict "purity" of the AI agents' behaviors. Because these agents operate based on foundational prompts and parameters set by their human creators, the observed social dynamics might not represent purely autonomous behavior; emergent network structures may be confounded by prompt engineering biases. This limitation will be addressed transparently in the final discussion.

The research will be executed within a school year.

May–September (completed): Finalized data ingestion of the 750K Hugging Face dataset. Conducted pilot analysis on a 10K-post subset, including Random Forest feature importance extraction, Multinomial Logistic Regression for class-specific engagement drivers, and preliminary social network visualization of top-centrality agents (see Section 4).

October–November: Deploy distributed computing architecture for full-scale processing. Execute BERTopic modeling and BERT-based sentiment/emotion extraction across the full

[2] https://github.com/macs-32000-26/replication-materials-JingwenJennyLU/tree/main

750K corpus. Train and evaluate Random Forest and Multinomial Logistic Regression models alongside gradient-boosted alternatives (XGBoost/LightGBM) using the extracted NLP features at full scale.

November–December: Generate the full directed agent interaction graph. Run community detection algorithms (Louvain/Leiden) and compute all node-level and network-level metrics outlined in Section 3.4. Execute null model validation suite: permutation tests on the interaction network, temporal shuffling, and topic-interaction decoupling analyses (see Section 3.5).

January: Synthesize analytical outputs, finalize data visualizations, and draft the concluding sections of the manuscript.

February: Submit the first draft of the thesis and revise according to faculty advisor's feedback.

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